

- (c) All three (d) None

I.A.S. (Pre) 2023

Ans. (b)

Native to India and Malaysia, Jackfruit tree is an evergreen tree and approx. 25 m tall. This tree produces latex and spreads to Sri Lanka, China, Southeast Asia and to tropical regions of Africa.

Significantly, Mahua is a tropical deciduous fast-growing tree and spreads to Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, Maharashtra etc. It is worth mentioning that Teak is a tropical hardwood tree. It is deciduous in nature.

29. One of the following regions has the world's largest tropical peatland, which holds about three years worth of global carbon emissions from fossil fuels; and the possible destruction of which can exert detrimental effect on the global climate. Which one of the following denotes that region?

- (a) Amazon Basin (b) Congo Basin
(c) Kikori Basin (d) Rio de la Plata Basin

I.A.S. (Pre) 2024

Ans. (b)

As per the article "Critical Ecosystems : Congo Basin Peatlands" published on the UNEP website in February, 2023, the Congo Basin is home to the world's largest tropical peatlands, along with Brazil and Indonesia. The peat swamp forest of the Congo Basin stores around 29 billion tons of carbon - approximately equivalent to three years' worth of global greenhouse gas emissions - while the Basin as a whole absorbs nearly 1.5 billion tons of carbon dioxide a year. Peatlands are an effective carbon sink - they absorb more carbon from the atmosphere than they produce. Carbon sinks are essential to combating the climate crisis and protecting planetary health. However, peatlands and other carbon sinks are already at risk of collapse due to climate change, nature and biodiversity loss, and pollution and waste. Without the critical services these ecosystems provide, the climate and nature crisis will only worsen.

The World Climate

*Climate is the average weather in a place over a certain period of time. It is determined by the daily weather conditions of a place (like temperature, rainfall, winds, humidity, air pressure etc) persisting over a long duration of time. * Earth has been basically divided into three climatic

zones – **Tropical Zone, Temperate Zone** and **Frigid Zone**.

These zones extend from the equatorial region towards poles.

* The continent of Asia extends from equatorial region to North Pole. As a result, all types of climates are found here. Different climates are found in different parts of the world. Places having same climates are called climatic regions and they are as follows :

• **Equatorial Climatic Region** – Found in between 0° to 10° North and South of the equator. The temperature remains high in this region all throughout the year. As a result, the region remains humid and it receives rainfall in the afternoon. The flora of the region consists of broadleaf evergreen trees like Gutta Percha, Rubber, Ebony, Palm, Bamboo, Bet, Mahogany, Sandal, Sinkona, etc. The regions under this climate include – Amazon Basin, the Congo Basin of Africa, Guinea Coast, Eastern island groups, and eastern Central American (Panama, Costa-Rica, Nicaragua, Honduras, Guatemala).

• **Tropical Monsoonal Climatic Region** – The region lies in the route of trade winds. Places included in this region are – **Pakistan, India, Bangladesh, Myanmar, Thailand, Cambodia, Laos, Eastern Coastal Regions of Africa, South-Eastern coastal parts of USA and northern parts of Australia** etc.

• **Sahara Type of Climatic Region** – Also known as a **tropical dry desert type** of climate. The climate extends between 15° to 30° latitudes on both sides of the equator in the western parts of the continent. There are two distinct seasons in this type of climate, the summer season and the winter season. Seasons are distinguished based on temperature. The average temperature of the summers is between 30°-35°C. According to the readings of NASA's satellite (2003-2009) **Dasht-e-Lut** desert of Iran was the hottest place on the planet based on surface temperature. Highest recorded temperature in 2005 was 70.7°C. Places under Sahara type of climate includes – **Thar desert and Arabian Peninsula of Asia, Sahara and Kalahari deserts of Africa, Western deserts of Australia, Atacama desert of South America and Western desert of North America**.

• **Savanna type of climatic region** – the temperature of the equatorial type of climate and savanna type of climate are somewhat the same. Since the savanna type of climate lies between the equatorial type of climate and tropical desert climate, the region is dominated by grasslands. The savanna type of climate is also called **Sudan type** of climate. There is a distinct dry

and moist season however the temperature is high throughout the year. The places included in this region are – **Venezuela, Colombia, Guyana, South-Central Brazil, and Paraguay.** The maximum extent of savanna type of climate in Africa is found in Sudan and northern region of Australia.

• **Mediterranean Type of Climatic Region :** Also known as sub-tropical dry summer type of climate. Since this type of climate is found near the Mediterranean Sea it is known as Mediterranean Climate. Mediterranean type of climate is found between 30° to 40° latitudes in both the hemispheres on the western parts of the continents. Maximum rainfall occurs in winter. Summers are dry and winters are wet. The summer winds are dry in nature. The places included under the Mediterranean type of climate are – river valleys of France, southern Italy, Greece, western Turkiye, western Israel, and North Western Africa which includes Algeria, Southern California of North America, Central Chile, and south-western part of South Africa.

• **Steppe Type of Climate** – This type of climate is found in the temperate grassland regions. This type of climate is found in the middle latitudes in the central parts of the continent. The annual range of temperature is high and annual rainfall is between 250 to 750 milimeter. Places included in steppe type of climate are – **grasslands of USA and Canada, grasslands of South Australia and Eurasia.**

• **Western European Climate** – Found between 40° to 65° latitudes in both the hemispheres in the western part of the continents. The average temperature is between 15°-21°C. There is a good amount of rainfall in this climate however rainfall increases in winter as compared to summer. Places included in Western European type of climate are – **North Western Europe, Western Norway, Denmark, North Western Germany, France, Oregon and Washington in the USA, British Columbia in Canada, Chile and New Zealand.**

• **St. Lawrence Type of Climate** – It is a moist continental type of climate, found on the eastern ends of the continent, predominantly in the Northern Hemisphere between **45° to 65°** latitudes. There is a good amount of rainfall throughout the year but more in summer as compared to winter. The places included under St. Lawrence type of climate are – parts of **Argentina, Manchuria region of China, St. Lawrence Valley of America and the coastal provinces of Canada.**

• **Taiga Type of Climate** – Also known as a Siberian type of climate which comes under microthermal climate. The

winters are long and the summers are of short duration. The 10°C temperature the warmest month of a region determines the Northern limit of Taiga. Winter season is approximately 8 months long during which the temperature remains well below freezing point. *Beyond the Polar regions lowest temperature was recorded in **Greenland** – 69.6°C (-93.3°F) in December, 1991. * The lowest temperature in the world was recorded in **Vostok** in Eastern Antarctica on 21st July, 1983, which was – 89.2°C (-128.6°F). The average annual rainfall in Taiga type of climate is between 370 to 600 millimeters, however, maximum rainfall is in the form of snow. Places included in this type of climate are **Alaska of USA, Northernmost parts of Canada, from Scandinavia in the west to Siberia in the east, in Eurasia.**

• **Tundra Type of Climate** – It has the least vegetation and are found in Polar and Arctic regions. The ecological productivity is minimal. In tundra type of climate, there is a deficiency in solar heat as a result the temperature remains very low throughout the year. The annual average rainfall is less than 400 mm. Tundras extend from the southern limits of northern glaciers to southern limits of Taiga in the South, and include the regions of America and Eurasia. Tundra only supports limited flora such as **Lichens and Moss Flora** that can tolerate cold polar climate for example, those of high mountains, Alpines and tundra region are called **Hekistotherms.**

1. **Variations in the length of day and night time from season to season are due to –**

1. **The earth's rotation on its axis**
2. **The earth's revolution around the sun in an elliptical manner.**
3. **Latitudinal position of the place.**
4. **Revolution of the earth on axis**

Which of the above-mentioned statement/statements is/are correct?

- | | |
|----------------|------------------|
| (a) 1 only | (b) 2 and 4 only |
| (c) 1, 2 and 3 | (d) 1 and 3 only |

I.A.S. (Pre) 2013

Ans. (b)

Earth is tilted on its axis. If Earth had not been tilted on its axis, then days and nights would have been equal everywhere. In the same way, if Earth had not been revolving around the sun, then there would always have been longer days and shorter nights in one hemisphere and longer nights and shorter days in the other hemisphere. But except the equatorial line, the difference in the length of days and nights is seen in different seasons. On the equatorial line, days and nights are always equal.

2. The annual range of temperature in the interior of the continents is high as compared to coastal areas. What is/are the reason/ reasons?

- (1) Thermal difference between land and water.
- (2) Variation in altitude between continents and oceans
- (3) Presence of strong winds in the interior.
- (4) Heavy rains in the interior as compared to coasts.

Select the correct answer using the code given below:

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2, 3 and 4

I.A.S. (Pre) 2013

Ans. (a)

The reason behind the high annual range of temperature in the interior of the continents is the thermal difference between land and water. The land gets heated rapidly through the insolation of the sun whereas water gets heated very slowly. Due to this, the weather of the coastal areas remains cold compared to the terrestrial regions. Hence, option (a) is the correct answer. However the other statements given in the codes are not true.

3. The main factor which determines the climate of any area is

- (a) Altitude
- (b) Latitude
- (c) Types of vegetation
- (d) Proximity to the sea

U.P.P.S.C. (GIC) 2010

Ans. (b)

Latitude is the main factor that determines the climate of any area. The major belts of the climate have been divided according to the different latitudes.

4. By which of the following is the maximum percentage of Earth covered ?

- (a) Arid Region
- (b) Semi-arid Region
- (c) Sub-humid land
- (d) Humid Region

U.P. Lower Sub. (Pre) 2002

Ans. (a)

About one-third of the Earth's area is covered by arid regions which is greater than all other geographical regions.

5. Given below are two statements one labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A) : Areas lying within five to eight-degree latitude on either side of the equator receive rainfall throughout the year.

Reason (R) : High temperature and high humidity cause convectional rainfall mostly in the noon near the equator.

By using the codes given below, select the correct answer:
Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2003

I.A.S. (Pre) 1996

Ans. (a)

The climatic region of equatorial line spreads between 0°-10° latitudes in both hemispheres. There are two important features of this climatic region :

- (1) Constant high temperature throughout the year.
- (2) Constant high rainfall throughout the year. Hence, assertion (A) is correct .

Due to high temperature throughout the year, there is high humidity which results in convectional rains mostly after noon. As the day progresses, the intensity of rain gets reduced. Hence, (R) is also correct and (R) is the correct explanation of (A).

6. "Each day is more or less the same, the morning is clear and bright with a sea breeze; as the Sun climbs high in the sky, heat mounts up, dark clouds form, then the rain comes with thunder and lighting. But the rain is soon over."

Which of the following regions is described in the above passage?

- (a) Savannah
- (b) Equatorial
- (c) Monsoon
- (d) Mediterranean

I.A.S. (Pre) 2015

Ans. (b)

See the explanation of the above question.

7. Which of the following continents has all types of climatic zones?

- (a) South America
- (b) North America
- (c) Australia
- (d) Asia

U.P.P.C.S. (Mains) 2016

Ans. (d)

Earth is divided into three types of climatic zones – tropical, temperate and cold. These spread from the equatorial line to the poles. Asia extends from the equatorial region to the North Pole, hence it is home to all types of climatic zones.

8. Which among the following countries doesn't come under the category of equatorial climate ?

- (a) Congo
- (b) Ethiopia
- (c) Gabon
- (d) Zaire

U.P.U.D.A./L.D.A. (Pre) 2013

Ans. (b)

Ethiopia does not come under the category of equatorial climate. Congo (Republic of the Congo), Gabon and Zaire (Democratic Republic of the Congo) are situated on the equatorial line.

9. A geographic region has the following distinct characteristics :

1. Warm and dry climate.
2. Mild and wet winter.
3. Evergreen oak trees.

The above features are the distinct characteristics of which one of the following regions?

- (a) Mediterranean
- (b) Eastern China
- (c) Central Asia
- (d) Atlantic coast of North America

I.A.S. (Pre) 2010

Ans. (a)

Characteristics of a warm and dry climate, mild and wet winter and evergreen oak trees are related to the Mediterranean region. Hence, option (a) is the correct answer.

10. Which one of the following is the characteristic climate of the Tropical Savannah Region?

- (a) Rainfall throughout the year
- (b) Rainfall in winter only
- (c) An extremely short dry Season
- (d) A definite dry and wet Season

I.A.S. (Pre) 2012

Ans. (d)

Tropical Savannah region's climatic features are – clear dry and wet season, high temperatures throughout the year and high insolation. Average annual rainfall in this region is between 500 mm and 2000 mm and no month of the year has a temperature less than 20°C.

11. Assertion (A) : The environment of the equatorial regions is conducive for plants not for humans.

Reason (R) : The average annual temperature difference is very low in equatorial regions.

By using the code given below, select the correct answer:

- (a) Both (A) and (R) are individually true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are individually true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

U.P.U.D.A./L.D.A. (Spl) (Mains) 2010

Ans. (b)

The region between 0° to 10° latitude north and south of the Equator is known as the equatorial region. In this region, there is constant high temperature and heavy rainfall occurs throughout the year. The climate of this region is not conducive for humans but it is conducive for vegetation. Evergreen forests are found here. Humidity persists all day and hence the area becomes prone to disease.

12. Mediterranean climate is not found in which among the following town ?

- (a) Los Angeles
- (b) Rome
- (c) Cape town
- (d) New York

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (d)

Mediterranean climate extends 30°-40° latitudes around the Mediterranean Sea and is experienced in the western parts of continents besides Rome.

These areas are –

South California, Los Angeles in North America, Central Chile in South America, Cape Town in South Africa & South Australia..

Thus, it is clear that New York, located on the east coast of the USA does not fall under the Mediterranean climate.

13. Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A) : Rainfall occurs in the winter season in Mediterranean climate.

Reason (R) : In summer these regions remain under the influence of dry terrestrial winds.

Select the correct answer from the code given below :
Code :

- (a) (A) is true but (R) is false
- (b) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (c) (A) is false but (R) is true
- (d) Both (A) and (R) are true but (R) is not the correct explanation of (A)

U.P.P.C.S. (Pre) 2022

Ans. (d)

The Mediterranean climate has been developed between 30°-40° latitudes in both hemispheres. The Mediterranean type of climate is characterized by dry summers and wet winters. The basic cause of this type of climate is the shifting of pressure and wind belts.

In summer, when Sun is overhead at Tropic of Cancer, the Westerlies belt is shifted poleward. The prevailing Trade winds are off-shore (terrestrial) and there is 'no rain'. The Mediterranean lands receive most of their precipitation in winter when the westerlies shift equatorward. The westerlies are associated with temperate cyclones and blow from sea surfaces, hence being moist yields sufficient rainfall in coastal areas.

14. Consider the following statements with reference to Mediterranean climate:

1. It receives rainfall in winter season.
2. Due to winter solstice, air pressure belts shift to the South.

Which of the above statements is/are correct?

- (a) Only 2 (b) Neither 1 nor 2
(c) Only 1 (d) Both 1 and 2

UPPSC (Pre) 2024

Ans. (d)

The Mediterranean climate generally develops between 30° and 40° Latitudes in both hemispheres, primarily on the western sides of continents. A key characteristic of the Mediterranean climate is that rainfall occurs in winter, while summers are usually dry.

The development of the Mediterranean climate occurs due to the shifting of pressure and wind belts caused by the northward and southward movement of the sun (Uttarayan and Dakshinayan) During winter (in the Northern Hemisphere), at the time of the winter solstice when sun is over the Tropic of Capricorn, the westerly winds originating over the ocean expand into these northern latitudes. Since these moisture wind are associated with temperate cyclones, they cause rainfall.

During winter (in the context of the Southern Hemisphere), at the time of the summer solstice when the sun is over the tropic of cancer, the expansion of westerly winds originating over the oceans into these latitude bring winter rainfall.

Note : Statement (2) is technically incorrect. It is only valid for the mediterranean climate in the Northern Hemisphere. The development of the mediterranean climate is not solely due to the southward shift of pressure belts during the winter solstice but also due to the northward shift to Pressure belts during the summer solstice (which corresponds to winter in the southern Hemisphere).

15. "Climate is extreme, rainfall is scanty and the people used to be nomadic herders." The above statement best describes which of the following regions?

- (a) African Savannah
(b) Central Asian Steppe
(c) North American Prairie
(d) Siberian Tundra

I.A.S. (Pre) 2013

Ans. (b)

In Central Asian Steppes, the climate is extreme, rainfall is scanty and the people used to be nomadic herders.

16. Which of the following pair matchings is correct?

- (a) Mediterranean Sea Zone-Summer rain
(b) Equatorial Zone-rain with thunder in the noon
(c) Monsoon Zone-Heavy rain throughout the year
(d) Desert zone- winter rain

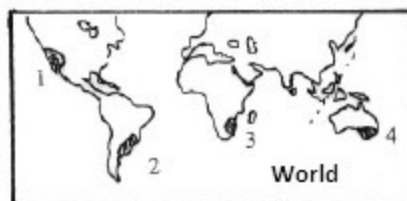
39th B.P.S.C. (Pre) 1994

Ans. (b)

The Mediterranean Sea climatic region receives rainfall in winter.

In equatorial regions, convectional rain occurs with thunder at noon almost throughout the year. In monsoon regions, the major rainfall is received from monsoon winds. In desert regions, rainfall is either negligible or indefinite. Hence, the correctly matched pair is option (b).

17. Consider the map given below :



Of the four shaded areas in the map, that which is characterized by hot dry summer, mild and moist winters and seasonal reversal of winds is the areas labelled.

- (a) 1 (b) 2

(c) 3

(d) 4

I.A.S. (Pre) 1997

Ans. (a)

Characteristics mentioned above such as hot, dry summers, mild and moist winter and seasonal reversal of winds describe the geographic characteristics of the Mediterranean climate. California (USA) labelled as 1 on the above map is in the Mediterranean region which has all the characteristics mentioned above.

18. In which among the following countries, rainfall occurs in the winter season?

- (a) China
- (b) Norway
- (c) Brazil
- (d) Nigeria

R.A.S./R.T.S. (Pre) 1993

Ans. (b)

Norway comes under the Western European type of climate (Western climate of both the hemispheres in continents between 40° -65° latitudes). In this climatic region, rainfall occurs in sufficient and equal quantities throughout the year. Whereas China comes under the China type climate although here also rains occur throughout the year winter season is relatively dry. Hence, option (b) is the correct answer.

19. Which one of the following is correctly matched ?

- (a) Wet winter dry summer - Italy
- (b) Wet throughout the year - Sudan
- (c) Wet summer dry winter - Iran
- (d) Dry throughout the year - Chile

U.P.P.C.S. (Mains) 2004

Ans. (a)

Most of Italy has a Mediterranean type of climate, which has, rainy winters and hot, dry summers. Sudan has a tropical climate. Summer temperatures often exceed 43°C in the desert zones, and rainfall is negligible. Iran has a hot, dry climate characterized by long, hot, dry summers and short winters. Northern Chile, situated in the desert zone, is very hot and dry. Very cold and dry during winters while western middle Chile has a Mediterranean type of climate. Hence, option (a) is the correct answer.

20. Match List-I with List-II and select the correct answer using the codes given below :

List-I

(Region)

- A. California
- B. West Australia
- C. Bangladesh
- D. Siberia

Code:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 1 | 4 | 2 | 3 |
| (c) | 2 | 3 | 4 | 1 |
| (d) | 2 | 4 | 1 | 3 |

List-II

(Type of Climate)

- 1. Mediterranean
- 2. Tropical Monsoon
- 3. Cold Temperate
- 4. Hot Desert

U.P.P.C.S. (Pre) 1999

Ans. (b)

The correctly matched regions with the related type of climate are as follows :

(Region)	(Type of Climate)
California (U. S. A.)	- Mediterranean
West Australia	- Hot Desert
Bangladesh	- Tropical Monsoon
Siberia (Russia)	- Cold Temperate
Hence, the correct code is option (b).	

21. Match List-I with List-II and select the correct answer from the codes given below :

List-I

(Type of climate)

- A. Equatorial
- B. Mediterranean
- C. Monsoon
- D. Savannah

List-II

(Region)

- 1. California
- 2. Bangladesh
- 3. Sudan
- 4. Congo

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 1 | 2 | 3 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 1 | 2 | 3 | 4 |
| (d) | 4 | 3 | 2 | 1 |

U.P.P.C.S. (Pre) 2003

U.P.U.D.A./L.D.A. (Pre) 2002

Ans. (a)

Congo (Republic of the Congo) is situated in Central Africa. Hence the climate is equatorial as the equatorial line passes through it.

California is situated in the western part of the United States of America. This is among the Mediterranean climatic regions.

Bangladesh is situated in South Asia and comes under the area of monsoon type of climate.

Savannah type of climate is found in Sudan,

Savannah type climate is also known as Sudan type climate. Hence, option (a) is the correct answer.

22. Consider the following statements:

1. In Equatorial regions, the year is divided into four main seasons.
2. In the Mediterranean region, summer receives more rain.
3. In China type climate, rainfall occurs throughout the year.
4. Tropical highlands exhibit vertical zonation of different climates.

Which of these statements are correct?

- (a) 1, 2, 3 and 4 (b) 1, 2 and 3
(c) 1, 2 and 4 (d) 3 and 4

I.A.S. (Pre) 2002

Ans. (d)

In Equatorial regions, there are mainly two seasons – Summer and winter. Hence, statement (1) is incorrect. In Mediterranean region, winter receives more rain. Hence, statement (2) is also incorrect. China type climate is extended between 25°-40° latitudes on the eastern part of the continent. Here, rainfall occurs throughout the year. Hence, statement (3) is correct. Tropical highlands exhibit vertical zonation of different climates. Hence, statement (4) is correct.

23. Assertion (A) : The Mediterranean regions have rainy winters

Reason (R) : During winter they lie in the belt of the Westerlies.

Select the correct answer from the code given below :

Code :

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)

(c) (A) is true, but (R) is false

(d) (A) is false, but (R) is true

U.P.P.C.S. (Mains) 2004

Ans. (a)

Mediterranean climate is caused by seasonal reversal. During summer, as Sun is in summer solstice position, all pressure belts shift towards the north. This results in expansion of sub tropical high pressure over this region. In the winter, when the Sun's position is in winter solstice, westerly winds extend over this region bringing cyclones from mid latitudes that cause rain.

24. When do Mediterranean regions receive heavy rainfall ?

- (a) Summer (b) Spring
(c) Winter (d) None of these

M.P.P.C.S. (Pre) 1999

Ans. (c)

See the explanation of the above question.

25. Assertion (A) : The ports of north west Europe are open throughout the year.

Reason (R) : The south-westerly winds blow over north-west Europe throughout the year.

Select the correct answer using the code given below.

Code :

- (a) Both (A) and (R) are true, and (R) is the correct the explanation of (A)
(b) Both (A) and but (R) are true, and (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Mains) 2002

Ans. (b)

Oceanic streams and prevalent winds have a considerable impact on the temperature of western European climatic regions. The expansion of Europe is in the belt of westerly winds. Most of the time winds blow from south-west. In Europe, none of the mountains is spread from north to south. Because there is no hurdle in the path, these winds make the temperature mild. The hot water of the North Atlantic flow does not allow the waters of nearby oceans of Western Europe to freeze. Due to this, the ports are open throughout the year. There is good rainfall in these regions due to westerly winds. That's why both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

26. Consider the following statements with respect to the climate of Western Europe:

1. Western Europe receives rainfall in all months.
 2. Western Europe is located in the belt of Westerlies.
- Which of the above statements is/are correct?

- (a) Both 1 and 2 (b) Neither 1 nor 2
(c) Only 1 (d) Only 2

UPPSC (Pre) 2024

Ans. (a)

The Western European type climate (Climate of Western Europe) is found between 40° and 65° latitudes along the western edges of continents within the westerlies belt. This climate is humid, with rainfall occurring throughout the year in all seasons. However, winter generally receives slightly more rainfall than summer. There is variation in the amount of rainfall from one place to another. Generally, the amount of rainfall decreases as one moves from the coast towards the interior regions.

Note : The climate of Western Europe is here taken to mean the Western European type climate. However, Mediterranean climate conditions prevail in the southern part of Western Europe. In this context option (b) would be the correct answer. In its answer key, UPPSC has considered option (a) as the correct answer for this question.

27. Consider the following description :

1. Annual and daily range of temperatures is low.
2. Precipitation occurs throughout the year.
3. Precipitation varies between 50 cm-250 cm.

What is this type of climate?

- (a) Equatorial climate
(b) China type climate
(c) Humid subtropical climate
(d) Marine West coast climate

I.A.S. (Pre) 2024

Ans. (d)

The description given in the question is of Marine West Coast climate. Marine West Coast climate is located poleward from the Mediterranean climate on the west coast of the continents. The main areas are: Northwestern Europe, west coast of North America, north of California, southern Chile, southeastern Australia and New Zealand. Due to marine influence, the temperature is moderate and in winter, it is warmer than for its latitude. Generally, the mean temperature in summer months ranges from 15°-20°C and in winter it ranges from 4° to 10°C. The annual and daily ranges of temperature are small or low. Precipitation occurs throughout the year. Precipitation varies greatly from 50 to 250 cm.

28. The temperature and rainfall record of a meteorological station are given below:

Months	Temperature (0°C)	Rainfall (cm)
J	9.4	12.2
F	10.6	9.1
M	11.7	7.9
A	12.2	2.5
M	13.3	1.8
J	13.9	0.3
J	13.9	-
A	14.4	-
S	15.6	0.8
O	15.0	2.5
N	13.3	6.1
D	10.6	11.7

Average Temperature : 12.8°C

Average Rainfall : 54.9 cm per annum

Identify the region having the above climatic patterns from amongst the following:

- (a) Mediterranean region
(b) Monsoon region
(c) Steppe region
(d) North-west European region

I.A.S. (Pre) 2001

Ans. (a)

Data mentioned above is related to Mediterranean region. In this region total rainfall is received during winter and summer is mostly dry as shown in the above data by negligible rains during the months of June and July. In such type of region, the temperature is nearly the same throughout the year.

29. Which one of the following climates are associated with the shifting of the pressure belts ?

Select the correct answer from the code given below :

1. Equatorial climate
2. Mediterranean climate
3. Monsoon climate
4. Western European climate

Code :

- (a) 1 and 2
(b) 1 and 3
(c) 2 and 3
(d) 2 and 4

U.P.P.C.S. (Mains) 2010

Ans. (c)

The Mediterranean and monsoon climates are associated with the transfer of pressure belts. The Mediterranean climate regions are found in the western parts of continents between 30°- 40° latitude in both the hemispheres. Sub- tropical high-pressure belts are found in the middle of 30°-35° latitudes. Trade winds blow during summers in these latitudes whereas in winters these high pressure belts shift towards the equatorial line. There is an emergence of monsoon winds due to the shifting of atmospheric pressure belts.

30. Which one of the following is the correct sequence of the climatic regions while going from Zaire to the Netherlands?

1. Equatorial climate
2. Mediterranean climate
3. Hot desert climate
4. West European climate

Code :

- (a) 1, 3, 2, 4 (b) 1, 4, 2, 3
(c) 2, 3, 4, 1 (d) 3, 2, 1, 4

U.P.P.C.S. (Pre) 2002

Ans. (a)

Zaire (Democratic Republic of the Congo) is a central African country whereas Netherlands is a Western European country. The sequence of climatic regions while going from Zaire (D.R.C.) to Netherland is as follows :

1. Zaire falls in the equatorial region so first climate is equatorial.
2. Hot desert climate in Sahara western region.
3. Mediterranean type of climate in the Mediterranean Sea region.
4. Finally enters Netherlands where there is a Western European type of climate.

Thus (a) is the correct sequence of climates while going from Zaire to Netherlands.

31. The world's wettest place is :

- (a) Cherrapunji
- (b) Mawsynram
- (c) Singapore
- (d) Waialeale

U.P.U.D.A./L.D.A. (Pre) 2001

U.P.P.C.S. (Pre) 1997

Ans. (b)

Mawsynram (Meghalaya) is the wettest place on earth.

32. The coldest place on the earth is :

- (a) Halifax
- (b) Chicago
- (c) Siachin
- (d) Verkhoyansk

U.P. Lower Sub. (Pre) 2004

U.P. Lower Sub. (Spl) (Pre) 2004

U.P.P.C.S. (Pre) 1999

Ans. (d)

The coldest place in the world outside the Polar region was – When the question was asked Verkhoyansk (Siberia) (–67.8°C) (recorded in 1933)

The lowest temperature in the world was recorded in Vostok in Antarctica on 21st July, 1983, which was 89.2°C

33. Match correctly :

- | | |
|-----------------------|-----------------------|
| A. The hottest | 1. Chile |
| B. The coldest | 2. Cherrapunji |
| C. The wettest | 3. Antarctica |
| D. The driest | 4. Sahara |

A	B	C	D
(a) 1	2	3	4
(b) 4	3	2	1
(c) 2	3	1	4
(d) 3	2	4	1

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

The correct match is as follows :

The hottest	-	Sahara desert
The coldest	-	Antarctica
The wettest	-	Cherrapunji
The driest	-	Chile

34. For short-term climatic predictions, which one of the following events, detected in the last decade, is associated with occasional weak monsoon rains in the India sub-continent?

- (a) La Nina
- (b) Movement of Jet Streams
- (c) El-Nino and Southern Oscillations
- (d) Greenhouse effect on a global level

I.A.S. (Pre) 2002

Ans. (c)

Originally the term El Nino applied to an annual weak warm ocean current that ran southwards along the coast of Peru. It affects the Indian monsoon rains and causes drought-like situations. La Nina is the cooling of water in the equatorial eastern Pacific Ocean which is opposite to El Nino.

35. La Nina is suspected to have caused recent floods in Australia. How is La Nina different from El Nino?

- 1. La Nina is characterized by unusually cold ocean temperature in the equatorial Indian Ocean Whereas El Nino is characterized by unusually warm ocean temperature in the equatorial Pacific Ocean.**
- 2. El Nino has an adverse effect on south-west monsoon on India, but La Nina has no effect on monsoon climate.**

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2011

Ans. (d)

El-Nino is related to the eastern part of the Pacific Ocean which increases the temperature of the surface water while La Nina is related to the large pool of warm water in the western Pacific Ocean. El Nino has an adverse effect on south-west Indian monsoon whereas due to La Nina, on the western side of the Pacific Ocean, rainfall is generally heavier which activates the Indian summer monsoon as well. So both the statements are wrong.

36. El-Nino develops in the

- (a) Pacific Ocean
(b) Indian Ocean
(c) Mediterranean Ocean
(d) Atlantic Ocean

U.P.P.C.S. (Mains) 2015

Ans. (a)

See the explanation of the above question.

37. With reference to the “EL-NINO”, which of the following statement/s is/are correct?

- (1) El Nino involves appearance of warm current off the coast of Peru in eastern pacific.**
- (2) This warm current increase the temperature of water on the Peruvian coast by 10°C there by increasing the amount of Plankton in the sea.**

**Select the correct answer using the code given below-
Code :**

- (a) Only 1 (b) Both 1 and 2
(c) Only 2 (d) Neither 1 nor 2

U.P.P.C.S. (Pre) 2023

Ans. (a)

El-Nino is a complex weather system that appears once every three to seven years, bringing drought, floods and other weather extremes to different parts of the world. The system involves oceanic and atmospheric phenomena with the appearance of warm currents off the coast of Peru in the eastern Pacific and affects weather in many places including India. El-Nino is merely an extension of the warm equatorial current which gets replaced temporarily by cold Peruvian current or Humboldt current. This current increases the temperature of water on the Peruvian coast by 10°C. This results in: (i) the distortion of equatorial atmospheric circulation; (ii) irregularities in the evaporation of sea water; (iii) reduction in the amount of Planktons which further reduces the number of fish in the sea. Hence option(a) is the correct answer.

38. Where does the El Nino current flow?

- (a) The Pacific Ocean
(b) The Indian Ocean
(c) The Bay of Bengal
(d) More than one of the above

69th B.P.S.C. (Pre), 2023

Ans. (a)

See the explanation of the above question.

39. The Southern Oscillation is a seesaw pattern between which of these two oceans?

- (a) Pacific and Atlantic Ocean
(b) Pacific and Indian Ocean
(c) Atlantic and Indian Ocean
(d) None of the above.

M.P.P.C.S. (Pre) 2024

Ans. (*)

The Southern Oscillation is defined as a pattern of atmospheric pressure variation (oscillation pattern) between the western and central eastern tropical pacific ocean. The Southern Oscillation, when associated with El Nino, is referred to as ENSO (El Nino-Southern Oscillation). Hence, it is clear that the Southern Oscillation is only associated with the pacific ocean. Therefore, option (d) is the correct answer. However, some authentic sources clarify that the Southern Oscillation is defined as a pattern of atmospheric pressure variation (oscillation pattern) between the oceans around Indonesia and Australia and the central and eastern

tropical Pacific Ocean. Southern Oscillation Index (SOI) is based on pressure data between Tahiti (located in the central Pacific Ocean, French Polynesia) and Port Darwin (in the Northern Territory of Australia).

Authentic sources clarify that Port Darwin is located on the coast of the Timor Sea, which is part of the Indian Ocean (according to Savindra Singh, Port Darwin is located on the coast of the Pacific Ocean). From the above description, it is clear that the Southern Oscillation is an air pressure oscillation pattern between the Pacific Ocean and the Indian Ocean. Therefore, option (b) can also be the correct answer, as indicated by the commission.

40. Select the incorrect statement from the following :

- (a) El Nino and La Nina are a kind of storm.
- (b) During El Nino, there are fewer phytoplanktons off the coast.
- (c) During La Nina trade winds are stronger than usual.
- (d) El Nino & La Nina are the climate patterns.

Raj. P.C.S. (Pre) 2024

Ans. (a)

El Nino and La Nina are weather patterns. El Nino is associated with an increase in the temperature of the central and eastern tropical Pacific Ocean water, while La Nina is related to the cooling of the western Pacific Ocean water. During La Nina, trade winds become stronger than usual, pushing more warm water toward Asia. Therefore, El Nino and La Nina are not storms.

During El Nino, the process of upwelling in the central and eastern tropical Pacific Ocean slows down or stops, leading to a decrease in phytoplankton off the coast.

41. Analyse the following statistical data of temperature and rainfall :

Month	Temperature (°C)	Rainfall (cm)
January	6.7	14.0
February	6.7	13.2
March	7.2	11.4
April	8.9	11.4
May	11.1	8.1
June	13.9	8.1
July	15.0	9.6
August	15.0	12.2
September	13.9	10.4
October	11.1	14.4

November	8.9	14.0
December	7.8	16.8

The climate, to which this data is related to , is :

- (a) St. Lawrence Type
- (b) China type
- (c) Western European type
- (d) Mediterranean type

I.A.S. (Pre) 1999

Ans. (c)

The above-mentioned data is related to the Western European type of climate (Northern Hemisphere) . Its average temperature remains between 15°-21°C. Rainfall is in sufficient quantity in every season of the year but compared to winter, scanty rainfall is received during the summer season. In the above-mentioned data, less rainfall is shown in the months of May, June and July.

42. Consider the following climatic conditions (northern hemisphere) :

Month	Temp. °C	Rainfall Cm.
J	3.9	4.7
F	4.4	5.7
M	8.3	8.2
A	14.4	9.2
M	20.0	9.2
J	23.3	17.7
J	27.8	14.5
A	27.8	14.0
S	22.3	12.7
O	18.3	7.0
N	12.2	5.0
D	6.7	3.5

These are most likely to be found in the natural regions of :

- (a) China type
- (b) Equatorial type
- (c) Hot desert type
- (d) Monsoon type

I.A.S. (Pre) 1998

Ans. (a)

Normally rainfall occurs throughout the year in China type of climate region but it occurs more during the summer season. The average temperature of this climatic region during the summer season is between 24°- 26.6°C. Hence, in the above- mentioned climatic chart, the features of China type of climatic conditions are shown.